

MAHINDRA AUTOMOTIVE

MR CAR ONBOARDING

Experiential mastery through high-fidelity Mixed Reality passthrough.



THE INDUSTRY

AUTOMOTIVE EVOLUTION

The global automotive industry is moving from mechanical engineering to a tech-first Software Defined Vehicle (SDV) ecosystem.

LEGACY: Focus on horsepower and physical reliability.

THE SHIFT: Electrification and autonomous driving systems.

MODERN: User experience as the primary differentiator.





THE CHALLENGE

COMPLEXITY VS. CONFIDENCE

As vehicles become more intelligent, the "UX Deficit" in the showroom leads to customer hesitation.



THE PROBLEM:

High-tech features are invisible and hard to explain.

THE WHY:

Confused customers delay purchases, impacting conversion.

OUR APPROACH

DESIGN METHODOLOGY

01. RESEARCH

Mapping the gap between feature density and spatial understanding.

02. STORYBOARDING

Visualizing the high-friction "As-Is" vs. the seamless "To-Be" journey.

03. PIVOT

Choosing MR Passthrough over VR to maintain environment connection.

04. PROTOTYPE

Engineering spatial anchors and gated logic flows.

The solution

Mixed Reality passthrough overlays contextual intelligence directly onto the physical vehicle cockpit.

STAKEHOLDERS

TARGET PERSONAS

THE USER: Desiring mastery over their ₹20L+ tech investment.

THE DEALERSHIP (THE BUYER): Needs tools to move aged inventory and secure deposits.

SALES STAFF: Requires frictionless, high-impact demonstration aids.



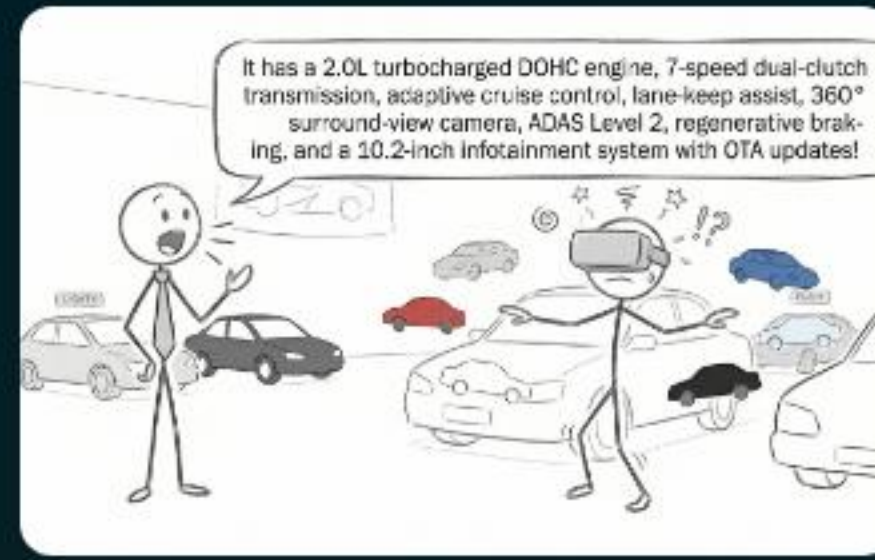
AS-IS JOURNEY

SHOWROOM FRICTION



ARRIVAL

Excitement meets technical apprehension.



VR WALL

Occluded VR causes environment disconnect.



THE LOST SALE

Overload leads to "I need to think about it."

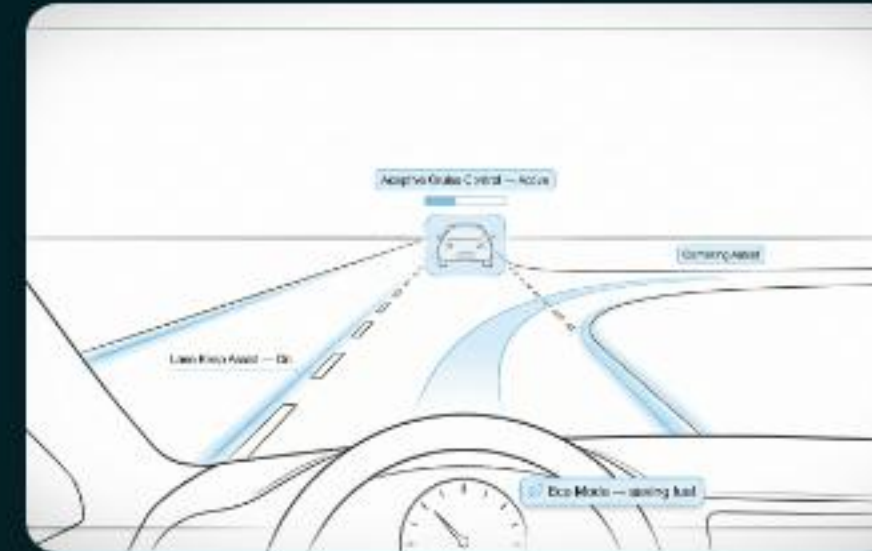
TO-BE RESOLUTION

THE MR INTERVENTION



THE PIVOT

Lightweight passthrough Visor.



DISCOVERY

Contextual digital signifiers.



CONFIDENCE

Learning through doing.

THE PIVOT

THE SIMULATOR HURDLE

VERSION CONTROL: OTA updates break 1:1 digital twins.

OS LOGIC: Rebuilding complex UI code is inefficient.

LIABILITY: Physics mismatches create sensory conflict.



SYSTEM DESIGN: FLOW 01

ENTRY & WELCOME

The initial MR handshake. A digital handshake between the showroom and the vehicle.

INTERACTION: "Start Level" CTA via pinch gesture.





SYSTEM DESIGN: FLOW 02

EXTERIOR DISCOVERY

Flush door handle spatial anchor with cyan outline and floating info dialog.

VOICE: "More info" / "Less info" state management.

SYSTEM DESIGN: FLOW 03

INTERIOR SETUP

Immediate tactile validation of vehicle intelligence.

SEQUENCE: Automatic seat sliding and height profile sync.





SYSTEM DESIGN: FLOW 04

AUDIO SYSTEM BRANCHING

The first experiential fork: User-led discovery of infotainment value.

DECISION: Experience vs. Skip offering user agency.



SYSTEM DESIGN: FLOW 05

PARALLEL TECH STACK

Unified anchor-highlight pattern across multiple cabin features.

FEATURES: AI Camera, Valet Mode, and Ambient Lighting.





SYSTEM DESIGN: FLOW 06

ADAS & GATED ACTIONS

High-confidence features gated by physical safety requirements.

GATE: Seatbelt fastened required to unlock Adaptive Cruise mastery.

LOGIC ARCHITECTURE

SYSTEM LOGIC FLOW

TRIGGERS: Proximity and physical button state.

NAVIGATION: Gesture-led chapter traversal.

Mahindra XEV 9e — MR onboarding user flow

Slides for AI integration reference

SLIDE 1 — Entry & welcome

User sits in XEV 9e
MR headset is active

Welcome dialog appears
"Car Onboarding Experience" + interaction hint

User selects input mode
Pinch gesture OR Voice: "Start Level"

SLIDE 2 — Exterior exploration

Spatial anchor: flush door handle
Cyan highlight outlines handle

Information dialog floats in space
Select for more info / voice: "More info"

User advances to interior
Pinch "Next Feature" or voice command

SLIDE 3 — Interior setup: seat & height profile

Feature: automatic seat sliding
Cyan outline on driver seat + info dialog

SPATIAL DESIGN

DESIGNING FOR THE EYE

Comfort-first UI placement to ensure prolonged engagement without fatigue.

FOCAL DEPTH: UI panels hover 1-2m from eyes to eliminate conflict.

WORLD-LOCKED: Persistent anchors pinned to dash surfaces.

CLARITY: High-contrast components calibrated for headset resolution.





OPERATIONAL LOGIC

10 MINUTES. THEN REST.

Designing around the physical limits of current-gen MR hardware.

SESSION CAP: 10–15 min window to avoid thermal discomfort.

PACING: Audio narrative enforces slow, deliberate exploration.

STABILITY: Design to discourage fast head movements and drift.

INTERACTION MODALITY

POINT. DON'T REACH.

Far-field ray-casting eliminates "gorilla arm" fatigue during onboarding.

MAGNETIC SNAPPING: Cursor gravity-wells to button centers.

HITBOXES: Oversized target zones to absorb hand tremor.

HOVER STATES: Dynamic scaling and audio hum for confirmation.



SAFETY OVERRIDE

DRIVE MODE = VOICE ONLY

Physical safety protocols that dynamically override interaction design.

LOCKOUT: Pinch gestures disabled instantly when car enters Drive.

HANDS-ON: No hand must leave the wheel during ADAS demos.

TRANSITION: Mode switch feels like a natural safety handoff.



SYSTEM READINESS

VOICE CALIBRATION

Ensuring multimodal reliability before high-speed maneuvers.



MICROPHONE CHECK

Initial trigger validation to confirm input gain.



COMMAND DIALOG

Floating global HUD showing active voice shortcuts.

SAFETY GATE

Voice interaction must be verified before the system allows the transition to Drive Mode.

THE CHALLENGE

SENSORY CONFLICT

When the Body Disagrees With the Eyes

Moving vehicles introduce the hardest ergonomic challenge in MR.

Sensory conflict: eyes see a stable digital overlay, inner ear feels the car turning — nausea begins within seconds.

Every design decision in this onboarding flow is built around resolving this single core tension.





MITIGATION 01

SPATIAL ANCHORING

The Dashboard Knows Where It Is

All MR overlays are world-locked to physical XEV 9e surfaces — dashboard, steering wheel, console.

Tracking is fused with car telemetry/IMU data; UI bounces in perfect sync with the physical dashboard.

Result: Zero drift between digital anchor and physical button, even at highway speeds.

MITIGATION 02

DEPTH CONSISTENCY

Digital depth must match physical depth

WINDSHIELD HUD: UI projected at road depth to eliminate eye strain.

STEERING TOOLTIP: UI projected at 0.5–0.7m, matching physical distance.

STABILITY: Onboarding flow never forces rapid depth toggling.

MITIGATION 03

CABIN UI DESIGN

Visual restraint is a safety feature, not a design choice

VIGNETTING: Peripheral elements fade during acceleration to reduce conflict.

TRANSPARENCY: High-transparency panels keep the brain spatially grounded.

PERFORMANCE: Strict < 20ms latency and 90Hz+ rendering.

MITIGATION 04

INFORMATION DIET

One Sentence. Three Seconds. Gone.

In Drive mode, overlays are a single sentence maximum and auto-fade within 4s.

Cognitive load reduction prevents nausea acceleration.

Pinching and ray-casting disabled; system relies on voice and telemetry.



SAFETY PROTOCOL

THE SAFE HANDOFF

The system adapts automatically using XEV 9e telemetry

PARKED

Full ray-cast UI. Voice + pinch active.
Rich info panels.

ENGAGED

Toast: "Voice Mode Active". Pinch
disabled. Hands on wheel.

IN MOTION

Telemetry-triggered only. 3s auto-fade.
Zero physical input.

SUMMARY

SAFE BY DESIGN

TRACKING: IMU + car telemetry fusion solves drift.

OPTICS: Depth-matched UI placement reduces strain.

VISUALS: Dynamic vignetting + high translucency.

LOAD: 3s overlays, voice-only in Drive mode.

"Designed to never make the user feel sick in the process."

NO SIMULATION
REQUIRED

SPATIAL ANCHORS — Digital Overlays

MR PASSTHROUGH VISOR — Video Feed

PHYSICAL CAR — Steering Wheel, Controls

DESIGN SYSTEM

UI STANDARDS

Built upon the Meta Horizon OS UI Set for proven spatial usability.

COLOR: Electric Cyan primary, Deep Violet accents.

TYPOGRAPHY: Archivo (Headings) / Inter (Body).

FIGMA: [Design Reference Page](#)

Experience the Camera features

You can experience the camera features from the dashboard or skip to next feature.

Experience

Next Feature

COMPONENT LIBRARY

INTERACTIVE CARDS

Precision-engineered "Half-State" units for initial focal discovery.

SIZE: 277px x 80px (Standard Entry footprint).

HITBOX: 48px minimum target zone for ray-cast stability.

SPACING: 24px left/top anchor padding.

MATERIAL: 168.5° linear-gradient glassmorphism.



Audio System



Audio System

Harman Kardon audio system. It is a 16-speaker, 1400W setup equipped with Dolby Atmos and 14 amplifiers, designed to provide an immersive "amphitheater" experience.

COMPONENT LIBRARY

INFO PANELS

High-density "Full-State" overlays for deep-dive feature explanation.

DIMENSIONS: 277px width / Content-adaptive height.

LAYOUT: Flex-column with 16px element gap.

TEXT: Semi-Bold 16px Headers / 14px Regular Body.

SHADOW: Multi-layered 60px soft white glow for Z-depth.



ADAS Vision Camera



- Adaptive Cruise Control.

COMPONENT LIBRARY

FEATURE CAROUSEL

Horizontal traversal for high-density feature discovery.

GESTURE: Hand-flick scrolling.

FOCUS: Center card scales 1.1x with descriptive text.

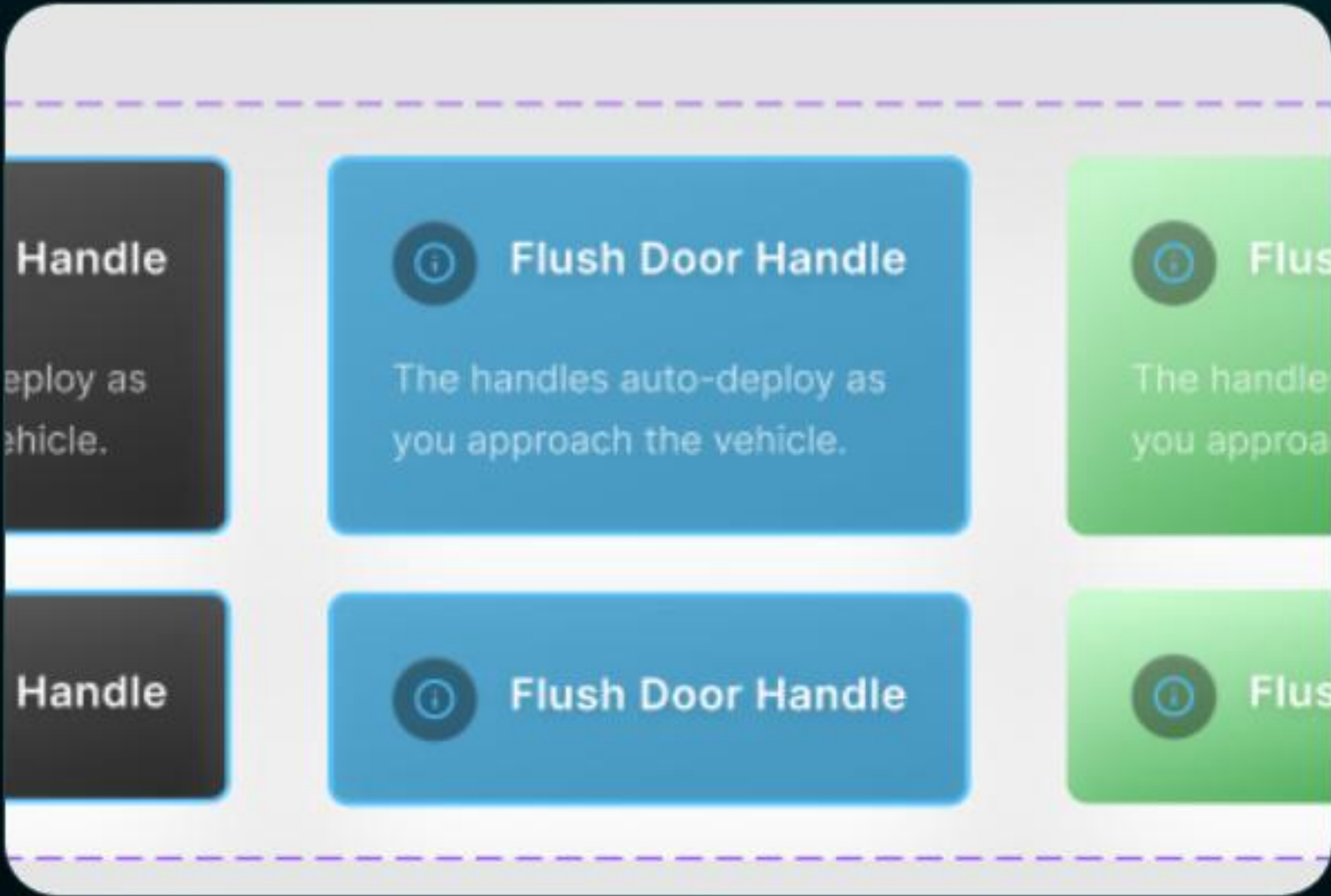
TRIGGERS: Ray-cast edge buttons or global voice commands.

USER FEEDBACK

STATE LOGIC

Real-time visual confirmation of user intent and spatial execution.

- HOVER:** Radiant border glow and subtle Z-axis scaling.
- PRESSED:** High-saturation Cyan fill with haptic "click".
- SUCCESS:** Teal (#0A9396) pulse on task completion.
- CONTRACT:** Dynamic scaling for detail focus management.



FEASIBILITY

CONSTRAINTS

GLARE & TRACKING

Showroom lighting management.

POWER & HYGIENE

Stationary uptime & sanitization.